Research Methodology

Module 2

Literature Review and Research Design

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1 Importance of Literature Review

1.1 What Is a Literature Review?

A literature review is a **critical study of existing knowledge** on a topic: books, journal articles, reports, and other scholarly sources. It is not just a list of summaries. It compares, connects, and evaluates earlier work to show *what is known, what is disputed, and what is still missing.*

> **Literature Review** A systematic reading, analysis, and written summary of earlier research related to a problem, used to place a new study within what is already known.

1.2 Why It Matters

- **Avoids duplication:** shows whether the question has already been answered.
- **Sharpens the problem:** helps narrow a vague idea into a clear research problem.
- **Builds the theoretical base:** identifies theories, concepts, and models to use.
- **Guides methodology:** shows which designs, samples, tools, and tests worked (or failed) before.
- **Reveals gaps:** points to unanswered questions and conflicting results.
- **Supports hypotheses:** gives evidence for predicting relationships between variables.
- **Helps interpret results:** later, findings can be compared with earlier studies.
- **Shows credibility:** proves the researcher knows the field.

1.3 Steps in Doing a Literature Review

1. Choose the topic and list keywords.
2. Search for sources in journals, databases, and libraries.
3. Screen them by relevance, date, and quality.
4. Read and take notes (author, year, aim, method, finding, limitation).
5. Organise by theme, method, or time, not source by source.
6. Critically evaluate, then write the review and cite everything.

> **Exam Tip** A good review is **thematic**. Instead of "Smith said X, then Jones said Y," write "Several studies report X (Smith, 2020; Rao, 2022), but Jones (2021) found the opposite."

> **Note** **Types of review:** *narrative* (general overview), *systematic* (strict, repeatable search and selection rules), and *meta-analysis* (statistically combines results of many studies).

2 Sources of Literature

Sources are usually grouped as **primary** (original research reports by the authors), **secondary** (reviews, textbooks that interpret primary work), and **tertiary** (encyclopaedias, indexes, directories).

2.1 Journals

Peer-reviewed journals are the **most trusted source**. Articles are checked by independent experts before publication.

- Contain original research, reviews, and case studies.
- Judged by indexing, impact factor, and reputation of the publisher.
- **Examples:** IEEE journals, ACM journals, Springer and Elsevier titles.

2.2 Conference Papers

Papers presented at academic conferences and published in proceedings.

- Report *recent* work faster than journals.
- Often shorter and less detailed; review depth varies by conference.
- Particularly important in computer science and engineering.

2.3 Reports

Documents from governments, institutions, industry bodies, and research organisations.

- Include technical reports, white papers, theses and dissertations, and statistical reports.
- Rich in data and real-world detail.
- May not be peer-reviewed, so check the author and purpose.

2.4 Online Databases

Searchable collections that index or host papers, books, and reports.

Database	Typical use
Google Scholar	Free, broad search across disciplines; shows citation counts.
IEEE Xplore	Computing and electronics journals and conferences.
ACM Digital Library	Computer science papers and proceedings.
Scopus / Web of Science	Large indexing databases to find and track quality journals.
JSTOR	Humanities, social sciences, and older journal archives.
ScienceDirect, SpringerLink	Full-text journals and books from major publishers.
Shodhganga	Indian theses and dissertations repository.

2.5 Judging the Quality of a Source

- **Authority:** Who wrote it, and from which institution?
- **Peer review:** Was it independently checked?
- **Currency:** Is it recent enough for the topic?

- **Relevance and accuracy:** Does it fit your question, with clear methods and evidence?
- **Objectivity:** Is there a hidden bias or commercial interest?

> **Watch Out** Blogs, social media, and unreferenced websites are **not** reliable academic sources. Use them only for ideas, then confirm from scholarly work.

3 Research Gap Identification

3.1 What Is a Research Gap?

A research gap is an **area that earlier studies have not studied, or have studied inadequately**. It is the reason a new study is worth doing.

3.2 Types of Gaps

Gap	Meaning	Example
Knowledge gap	Topic not studied enough.	No studies on app use in rural colleges.
Population gap	Studied for some groups only.	Only urban students sampled.
Methodological gap	Weak or limited methods used.	Only surveys, no experiments.
Contextual gap	Studied in another place or time.	Results from 2010, before smartphones.
Conflict gap	Studies disagree.	Some find a link, some find none.
Theoretical gap	Existing theory fails to explain.	Model ignores digital habits.

3.3 How to Find a Gap

1. Read recent reviews and look at the "limitations" and "future work" sections of papers.
2. Compare studies in a table: topic, sample, method, result.
3. Look for contradictions, missing groups, outdated data, and untested variables.
4. State the gap clearly in one or two sentences, then link it to your objectives.

> **Note** **Gap statement pattern:** "Although earlier studies have examined . . . , little is known about . . . in the context of . . . . This study therefore aims to . . ."

4 Types of Research Design

> **Research Design** The overall **plan or blueprint** of a study: what will be studied, on whom, with what methods, and how data will be collected and analysed to answer the research question.

A good design makes the study *valid* (measures what it should), *reliable* (consistent), *economical*, and *ethical*. The syllabus covers three types.

4.1 Experimental Design

The researcher **deliberately changes one variable** (independent) and observes its effect on another (dependent), while controlling other factors. It is the best design for showing *cause and effect*.

- Uses an **experimental group** (receives the treatment) and a **control group** (does not).
- Participants are usually assigned by **randomisation**.
- **Example:** Two groups of students; one uses a new study app for a month, the other does not. Compare marks.

Common forms:

- **True experimental:** random assignment, control group.
- **Quasi-experimental:** treatment given, but no random assignment (existing classes used).
- **Pre-experimental:** simple designs with weak control, such as a one-group before-and-after test.

> **Watch Out** Experiments give strong evidence for cause, but can be costly, artificial, and sometimes unethical or impossible with people.

4.2 Descriptive Design

Describes **what exists** without changing anything: characteristics, frequencies, and relationships in a population.

- Methods: surveys, observation, case studies, and fact-finding.
- **Cross-sectional:** data collected once, at one point in time.
- **Longitudinal:** the same group studied repeatedly over time.
- **Example:** A survey of internet usage hours among BCA students.

> **Note** Descriptive designs can show *that* two things are related, but they cannot prove that one *causes* the other.

4.3 Analytical Design

Goes beyond description to **explain why or how** something happens, by analysing existing facts or data, comparing groups, and testing relationships. The researcher does *not* manipulate the variables.

- Uses available data, records, or collected data analysed with statistics, logic, or critical evaluation.
- Includes **correlational**, **comparative**, and **case-control** studies.
- **Example:** Using last year's results to analyse whether attendance is related to marks.

4.4 Comparison

	Experimental	Descriptive	Analytical
Aim	Test cause and effect	Describe what exists	Explain relationships
Variables	Manipulated	Observed only	Observed, analysed
Control	High	Low	Moderate
Typical tools	Experiments, trials	Surveys, observation	Existing data, statistics
Example	New app vs. no app	Report app usage	Link attendance and marks

> **Exam Tip** Quick way to choose: *Want to prove cause?* Experimental. *Want a picture of the situation?* Descriptive. *Want to explain a relationship from existing data?* Analytical.

5 Developing Hypotheses and Conceptual Framework

5.1 Hypothesis

> **Hypothesis** A tentative, testable statement predicting a relationship between two or more variables, to be accepted or rejected after data analysis.

Characteristics of a good hypothesis: clear and specific, testable, based on literature or theory, states a relationship between variables, simple (one idea), and consistent with known facts.

Type	Meaning and example
Null (H_0)	No relationship or difference. *Study hours have no effect on marks.*
Alternative (H_1)	A relationship or difference exists. *Study hours affect marks.*
Directional	Predicts the direction. *More study hours raise marks.*
Non-directional	Predicts a link, not the direction. *Study hours are related to marks.*

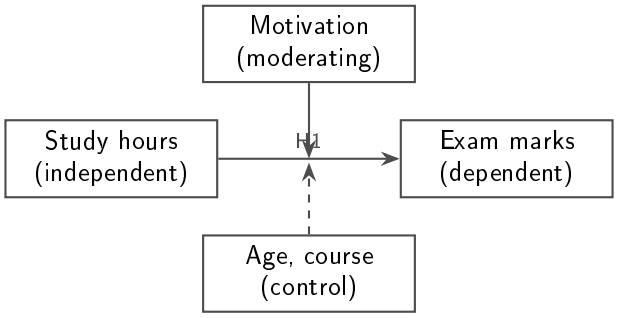
Steps in Developing a Hypothesis

1. Start from the research problem and the gap found in the literature.
2. Identify the independent and dependent variables.
3. Predict the relationship using theory and earlier findings.
4. Write it as a clear, testable statement (H_0 and H_1).
5. Check that the variables can actually be measured.

5.2 Conceptual Framework

Conceptual Framework A **visual or written model** showing the main variables of a study and the relationships the researcher expects between them, built from the literature.

It acts as the *map* of the study: it guides what to measure, which hypotheses to test, and how to analyse the data.



Steps to Build One

1. List the key concepts from your objectives and literature.
2. Define each as a measurable variable.
3. Decide which are independent, dependent, moderating, mediating, or control.
4. Draw arrows for the expected relationships and label each with its hypothesis.

	Theoretical framework	Conceptual framework
Source	Existing, established theory	Researcher's own model for this study
Scope	Broad and general	Specific to the study variables
Use	Explains why relationships exist	Shows what will be tested

Exam Tip Link everything: **literature** → **gap** → **objectives** → **conceptual framework** → **hypotheses** → **design**. Examiners like to see this chain.

6 Quick Revision Sheet

- **Literature review:** avoids duplication, sharpens the problem, guides method, reveals gaps.
- **Sources:** journals (most reliable), conference papers (recent), reports (data-rich), databases (search tools).
- **Research gap:** knowledge, population, methodological, contextual, conflict, theoretical.
- **Experimental:** manipulates a variable; shows cause and effect; control group.
- **Descriptive:** describes what exists; cross-sectional or longitudinal.
- **Analytical:** explains relationships from existing data without manipulation.
- **Hypothesis:** testable prediction; H_0 vs. H_1.
- **Conceptual framework:** map of variables and expected relationships.

End of Module 2